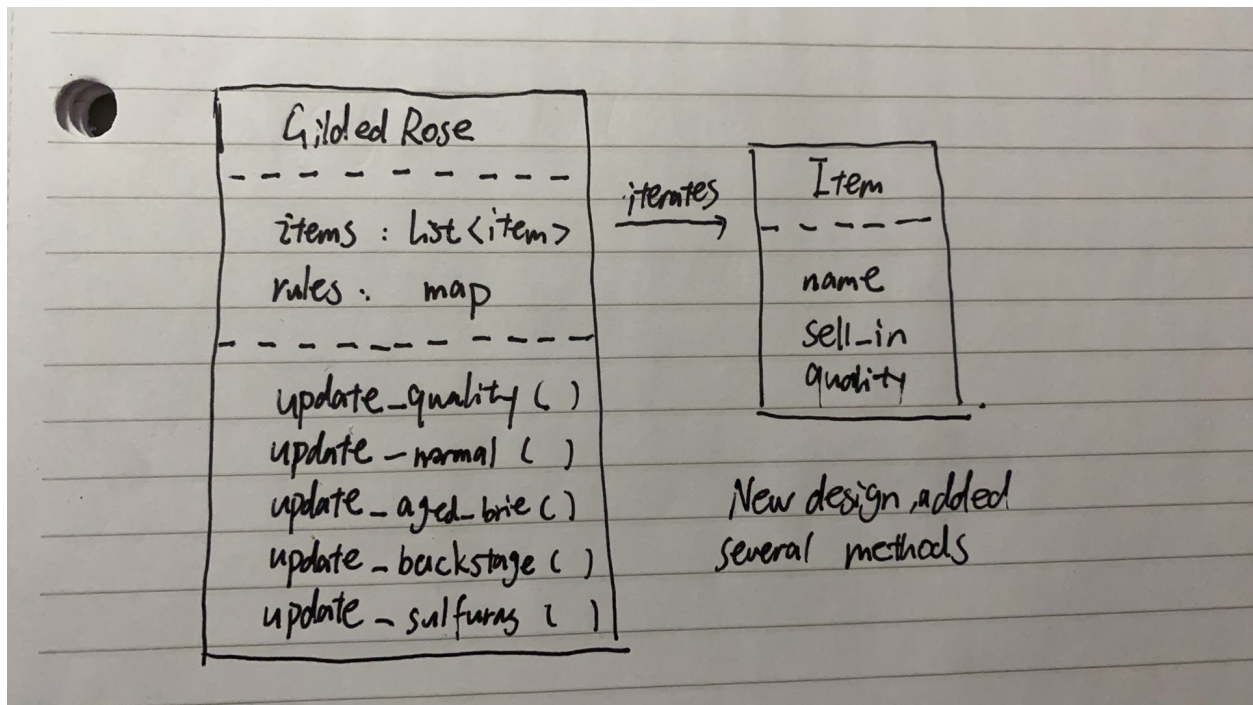




Major issues in the original code:

1. Nested if-else loop, repeated string check, very hard to read and understand.
2. Almost impossible to re-write the logic and new features or product
3. All rules are mixed in one method, so changing one might trigger a bug and collapsed all others. Difficult for testing.

My new Design:

GildedRose does not know the rules of each item. It only selects a strategy from the rules table and executes it. In my design, adding a new special item only requires implementing a new update strategy and registering it. The main update loop remains unchanged, which makes the system easier to extend and less error-prone. This design use the strategy design pattern in the behavioral patterns in the link <https://refactoring.guru/design-patterns/catalog>. Strategy pattern by mapping each item type to a separate update method. The main loop only selects and runs a strategy, which removes complex branching and improves clarity.